

YANGYANG INTL, Korea, Republic of

WMO: 470920

Lat: 38.061N

Lon: 128.669E

Elev: 241

StdP: 14.57

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 19.6	(d) -12.7	(e) 2.8	(f) 22.0	(g) -9.4	(h) 3.3	(i) 23.2	(j) 29.3	(k) 27.5	(l) 26.8	(m) 28.7	(n) 17.6	(o) 260	(p) 0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.2	(c) 89.2	(d) 75.0	(e) 86.2	(f) 74.1	(g) 84.1	(h) 72.5	(i) 77.7	(j) 83.9	(k) 76.4	(l) 82.8	(m) 75.2	(n) 81.8	(o) 9.8	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	135.1	80.2	74.8	131.7	79.8	73.3	124.8	78.8	41.2	83.3	40.0	82.7	38.8	82.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.3	(b) 21.9	(c) 19.2	(e) 11.5	(f) 93.6	(g) 4.5	(h) 2.1	(i) 8.3	(j) 95.0	(k) 5.7	(l) 96.3	(m) 3.2	(n) 97.4	(o) -0.1	(p) 98.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			8.0	78.8	4.0	1.7	5.1	80.0	2.7	80.9	0.5	81.9	-2.4	83.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	32.6	35.3	43.3	53.4	62.5	68.0	75.0	76.0	67.9	59.7	48.5	36.1
	DBStd	16.34	6.51	6.68	7.83	7.66	7.11	5.76	6.50	5.21	4.09	5.26	7.39	7.54
	HDD50	1793	541	413	239	47	1	0	0	0	0	4	115	433
	HDD65	4617	1006	832	672	357	135	30	1	0	15	176	496	897
	CDD50	3607	0	1	33	149	389	541	775	807	537	304	70	1
	CDD65	955	0	0	1	9	58	121	311	342	101	12	0	0
	CDH74	6216	0	0	1	46	310	616	2453	2536	239	15	0	0
	CDH80	1716	0	0	0	7	44	124	829	686	25	1	0	0
(14) Wind	WSAvg	8.4	11.3	9.9	9.7	9.3	8.1	5.8	6.1	6.1	6.1	7.6	9.6	11.5
(15) Precipitation	PrecAvg	58.0	1.8	1.9	2.3	2.8	4.0	5.2	10.8	12.0	9.2	3.4	3.0	1.5
	PrecMax	78.5	3.5	5.7	5.6	6.1	13.0	12.4	26.4	27.0	23.2	10.6	9.4	7.8
	PrecMin	43.5	0.1	0.1	0.2	0.7	0.6	0.9	4.0	2.6	1.0	0.4	0.2	0.0
	PrecStd	10.6	1.1	1.5	1.5	1.6	2.5	2.8	5.8	5.9	6.0	2.9	2.4	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	55.6	70.1	79.1	84.3	87.8	93.0	91.7	82.8	75.5	68.1	55.5
		MCWB	39.9	43.6	53.8	57.3	61.5	67.4	76.4	76.7	68.8	59.6	55.3	45.3
	2%	DB	46.6	50.4	64.3	73.4	80.6	82.8	89.6	88.2	79.0	71.9	64.5	51.9
		MCWB	38.2	40.7	50.0	54.1	61.0	65.8	75.4	75.9	68.1	59.4	53.2	44.2
	5%	DB	44.3	48.1	59.4	69.6	77.2	80.5	87.4	86.1	76.7	69.9	62.2	48.6
		MCWB	35.9	38.8	46.4	52.6	59.0	66.0	75.2	75.5	67.6	59.5	52.7	40.9
	10%	DB	41.3	44.9	55.5	65.8	74.8	77.2	84.4	84.0	74.8	68.0	59.1	46.4
		MCWB	33.7	36.7	43.5	50.8	58.0	64.9	73.7	74.0	67.1	58.8	50.5	39.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	45.9	55.6	59.5	66.5	72.0	79.1	79.4	73.2	67.2	59.4	49.3
		MCDB	47.2	52.0	67.9	74.1	78.7	80.2	85.7	85.1	77.6	69.2	63.2	52.8
	2%	WB	39.4	42.0	51.0	56.9	63.4	70.1	77.4	78.3	71.1	63.9	56.3	44.6
		MCDB	44.4	47.9	61.9	68.7	74.6	76.5	85.5	83.7	75.5	68.2	61.3	50.7
	5%	WB	36.9	40.0	47.8	54.7	61.6	68.9	76.0	77.1	69.7	62.1	54.5	41.8
		MCDB	41.9	45.7	56.7	64.8	71.7	74.8	84.3	82.3	74.1	66.9	60.0	47.2
	10%	WB	35.0	37.9	45.1	52.5	60.0	67.5	74.8	76.0	68.6	60.5	51.9	39.5
		MCDB	40.0	43.5	52.8	61.7	69.5	73.2	82.6	81.5	72.8	65.9	57.9	45.4
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.6	12.8	13.4	13.6	10.5	9.0	9.2	10.2	11.8	11.5	11.3
		MCDBR	12.3	14.6	17.4	18.7	18.4	16.0	12.5	11.9	12.4	13.5	12.6	12.1
		MCWBR	9.3	10.3	10.5	10.0	8.4	6.5	5.2	5.3	7.2	8.9	8.8	9.8
	5% WB	MCDBR	10.8	13.1	16.6	16.9	15.8	11.6	11.3	10.2	9.7	11.0	10.8	11.1
		MCWBR	9.0	10.1	11.1	10.4	8.7	6.0	5.6	5.9	6.6	8.5	8.6	9.9
(40) Clear-Sky Solar Irradiance	taub		0.409	0.471	0.562	0.608	0.614	0.662	0.628	0.555	0.485	0.480	0.445	0.401
	taud		1.995	1.870	1.722	1.677	1.713	1.677	1.813	1.999	2.119	1.996	2.014	2.043
	Ebn at Noon		229	230	221	221	224	213	219	233	242	225	215	221
	Edh at Noon		42	54	69	76	75	77	67	54	46	47	41	37
(44) All-Sky Solar Radiation	RadAvg		774	1019	1301	1558	1727	1640	1357	1349	1243	1080	762	686
	RadStd		69	101	128	127	171	133	188	170	149	92	102	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	+99

Nomenclature: See separate page